ECLT5930B & SEEM5740B: Engineering Economics

Sample Solution of Assignment 4

Instructions

This piece of assignment will count for 10 percent of the overall mark for ECLT5930B & SEEM5740B. (Four assignments in total. The best three will be counted.) There are 9 questions with a total of 100 marks. Do all of the questions.

Completed work should be submitted to the course website at Blackboard by ***7:00 pm on Monday 20 April 2026****. The deadline is strict. Work handed in late will not be accepted. All of the assignment must be carried out and written up independently. You are reminded of the University policy on plagiarism:*
`http://www.cuhk.edu.hk/policy/academichonesty/Eng_htm_files_(2013-14)/index_page2.htm`.
If you have any questions, feel free to contact yuantao@link.cuhk.edu.hk.

1. (8 marks) Consider the question on slide 20, lecture notes of Topic 8. Suppose that you handle the two most sensitive factors in a project: initial investment (I) and annual profit (A). PW is a function of the two variables I and A:

$$PW = -I + A(P/A, 10\%, 6) + 1000(P/F, 10\%, 6).$$

Suppose that the initial investment is restricted to the range $[11000, 14000]$ and the annual profit is expected to be in the range $[2400, 3100]$. Use the graphical method to estimate the best and the worst PW values.

Solution. The PW can be calculated as

$$PW = -I + 4.3553A + 564.$$

Since the PW is increase with the annual profit and decrease with the initial investment, then the best PW can be achieved with minimum initial investment $I = 11000$ and maximum annual profit $A = 3100$. In this case, PW will be

$$PW = -11000 + 4.3553 \times 3100 + 564 = 3065.43.$$

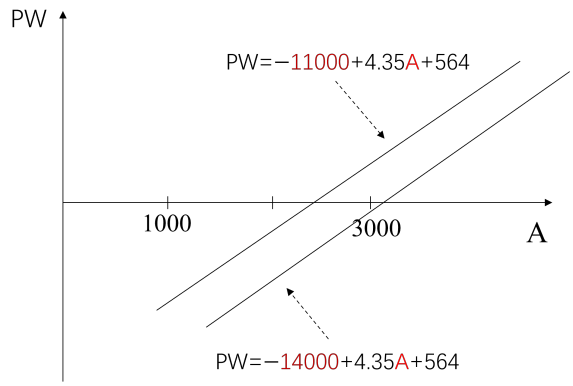

Figure 1: Q1-answer

Similarly, the worst PW can be achieved with maximum initial investment $I = 14000$ and minimum annual profit $A = 2400$. In this case, PW will be

$$PW = -14000 + 4.3553 \times 2400 + 564 = -2983.$$

2. (12 marks) A project requires an initial investment of \$100,000 and has a five-year study period. Assume that the annual cash flows are independent and identically distributed, and follows a normal distribution with expectation $\mu = 30000$ and variance $\sigma = 4000^2$. There is no salvage value at the end of year 5. The MARR is 10% per year.

   (a) (6 marks) Calculate the expected present worth, $E[PW]$, and the variance of the present worth, $\text{Var}(PW)$.

   (b) (6 marks) Consider another investment project. It requires an initial investment of \$90,000 and has a five-year study period. Assume that the annual cash flows are independent and identically distributed, and follows a normal distribution with expectation $\mu = 26000$ and variance $\sigma = 3000^2$. There is no salvage value at the end of year 5. Calculate the probability that the PW of this project is greater than the present worth of Project 1.

Solution. Part (a). The present worth of the project is

$$PW = -100{,}000 + X(P/A, 10\%, 5),$$

where $E[X] = 30{,}000$. Thus

$$E[PW] = -100{,}000 + 30{,}000\,(P/A, 10\%, 5) = 13{,}724.$$

Because the annual net cash flows are independent,

$$\begin{aligned}\mathrm{Var}(PW) &= \sum_{t=1}^{5}\left(\frac{1}{1.1^t}\right)^2 \mathrm{Var}(X) = 16{,}000{,}000\sum_{t=1}^{5}\frac{1}{1.21^t} \\ &\approx 16{,}000{,}000 \times 2.9260 = 46{,}815{,}749.\end{aligned}$$

Part (b). Similarly, we calculate the expectation and variance of project 2 are

$$\begin{aligned}E[PW_2] &= -90{,}000 + 26{,}000\,(P/A, 10\%, 5) \approx 8{,}560.46, \\ \mathrm{Var}(PW) &= \sum_{t=1}^{5}\left(\frac{1}{1.1^t}\right)^2 \mathrm{Var}(X) \approx 9{,}000{,}000 \times 2.9260 = 26{,}333{,}859.\end{aligned}$$

From part (a), for project 1,

$$E[PW_1] \approx 13{,}724, \qquad \mathrm{Var}(PW_1) \approx 46{,}815{,}749.$$

Because PW_1 and PW_2 are independent normal random variables,

$$\Delta(PW_2 - PW_1) \sim \mathcal{N}(\mu_\Delta, \sigma_\Delta^2),$$

where

$$\mu_\Delta = 8{,}560.46 - 13{,}723.60 = -5{,}163.15,$$

and

$$\sigma_\Delta^2 = 26{,}333{,}859.02 + 46{,}815{,}749.38 = 73{,}149{,}608.40.$$

Therefore,

$$\begin{aligned}P(PW_2 > PW_1) &= P(\Delta(PW_2 - PW_1) > 0) = 1 - \Phi\left(\frac{0 - (-5{,}163.15)}{\sqrt{73{,}149{,}608.40}}\right) \\ &\approx 1 - \Phi(0.6037) \approx 0.2729.\end{aligned}$$

Hence, the probability that Project 2 has a larger present worth than Project 1 is 27.29%.

3. (12 marks) Sabino County is considering whether to construct a culvert to reduce mudslide damage. There is a probability of 0.01 that one major mudslide will occur during that year. If a major mudslide occurs, the

county incurs a loss of \$1,000,000 at the end of that year because of reduced property-tax revenue. A civil engineer has proposed constructing a culvert on a mountain where mudslides are likely. This culvert will reduce the likelihood of a mudslide to zero. The investment cost would be $\$50,000$, and annual maintenance expenses would be \$2,000 in the first year, increasing by 5% per year thereafter. The culvert has no salvage value at the end of year 20. The county's cost of capital is 7% per year. Using the expected present worth criterion, answer the following questions:

(a) (8 marks) Assume that in each year there is an independent probability of 0.01 that one major mudslide will occur during that year. Should the culvert be built?

(b) (4 marks) Assume instead that over the 20-year analysis period, at most one major mudslide can occur. If no mudslide has occurred in earlier years, then the probability that a major mudslide occurs during year t is 0.01. If it occurs, the loss is incurred at the end of that year, and no further mudslide losses can occur later. Should the culvert be built?

Solution. Part (a). The expected mudslide loss in any year is

$$E[\text{annual loss}] = 0.01(\$1{,}000{,}000) = \$10{,}000.$$

Then, the expected present cost of doing nothing is

$$E[PC_{\text{DN}}] = 10{,}000(P/A, 7\%, 20) = 10{,}000(10.5940) = \$105{,}940.14.$$

If the culvert is built, then present worth of maintenance is

$$PW_{\text{MT}} = \sum_{t=1}^{20} \frac{2{,}000(1.05)^{t-1}}{(1.07)^t} = \frac{2{,}000}{1.07 - 1.05}\left[1 - \left(\frac{1.05}{1.07}\right)^{20}\right] = \$31{,}433.75.$$

Therefore, the expected present cost of building the culvert is

$$E[PC_{\text{B}}] = 50{,}000 + 31{,}433.75 = \$81{,}433.75.$$

Since

$$105{,}940.14 > 81{,}433.75,$$

, the county should build the culvert.

Part (b). The probability that the mudslide occurs in year t is

$$0.99^{t-1}(0.01), \qquad t = 1, 2, \ldots, 20.$$

Therefore, the expected present cost of doing nothing is

$$E[PC'_{\text{DN}}] = \sum_{t=1}^{20} \frac{1{,}000{,}000(0.01)(0.99)^{t-1}}{(1.07)^t} = 98{,}579.66.$$

Since

$$98{,}579.66 > 81{,}433.75,$$

the county should build the culvert.

4. (15 marks) A firm must decide between constructing a new facility or renting a comparable office space. There are two random outcomes for acquiring space, as shown in Figure 2. Each would accommodate the expected growth of this company over the next 10 years. The cost of rental space is expected to escalate over the 10 years for each rental outcome. The option of constructing a new facility is also defined in Figure 2. An initial facility could be constructed with the costs shown. In five years, additional space will be required. At that time, there will be an option to build an office addition or rent space for the additional space requirements. The probabilities for each alternative are shown. MARR for the situation is 10% per year. A PW analysis is to be conducted on the alternatives. Which course of action should be recommended?

   (Note: At Node 2 , the PW(10%) of the upper branch is -\$5,111.14 k and the PW(10%) for the lower branch is -\$4,119.06 k.)

   Solution. If the firm choose to build new facility at 1 (year 0), then it will choose to rent addition at 2 (EOY 5) since the PW of rent additional (-4119.06k at year 5) is less than that of build addition (-5111.14k at year

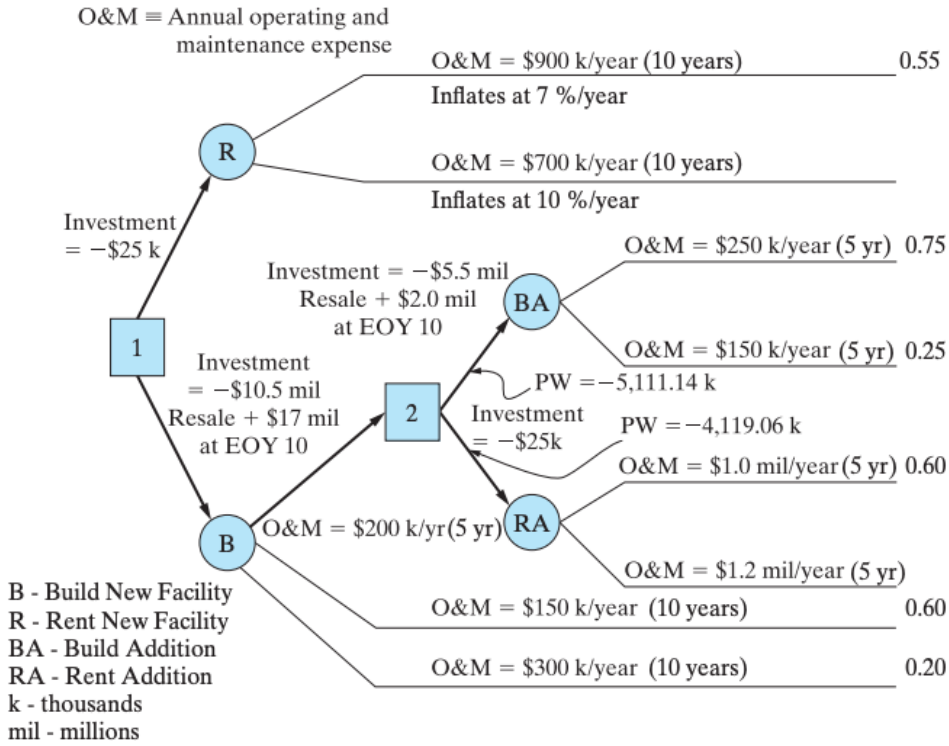

Figure 2: Question 4

5). The present worth of the cost of building new facility:

$$
\begin{aligned}
PW_B &= -10500 + 17000(P/F, 10\%, 10) \\
&\quad -0.2 * 300(P/A, 10\%, 10) - 0.6 * 150(P/A, 10\%, 10) \\
&\quad -0.2 * (200(P/A, 10\%, 5) + 4119.06(P/F, 10\%, 5)) \\
&= -10500 + 6554.23592 - 0.2 * 1,843.37 - 0.6 * 921.69 \\
&\quad -0.2 * (758.16 + 2557.57493) \\
&= -5530.6.
\end{aligned}
$$

The present worth of the cost of rent new facility:

$$\begin{aligned}
PW_R &= -25 - 0.55 * \sum_{i=1}^{10} 900 * (1+0.07)^{i-1}/(1+0.1)^i \\
&\quad -0.45 * \sum_{i=1}^{10} 700 * (1+0.1)^{i-1}/(1+0.1)^i \\
&= -25 - 0.55 * 900 * \frac{1-(1.07/1.1)^{10}}{1-1.07/1.1} * \frac{1}{1.1} \\
&\quad -0.45 * 700 * 10/1.1 \\
&= -6874.67.
\end{aligned}$$

Therefore, the firm should build new facility at node 1 and rent new facility at node 2.

5. (10 marks) Three industrial centrifuge designs, labeled A, B, and C, are being considered for a new chemical plant. Use the additive weighting method to determine which design should be adopted.

   Hint: use the following orderings for the qualitative attributes:

   Fair < Good < Excellent, Not known < Good < Excellent.

Table 1: Data for the centrifuge design alternatives

Attribute	Weight	A	B	C
Initial cost	0.25	\$140,000	\$180,000	\$100,000
Maintenance	0.10	Good	Excellent	Fair
Safety	0.15	Not known	Good	Excellent
Reliability	0.20	98%	99%	94%
Product quality	0.30	Good	Excellent	Good

Solution. For numerical attributes, the nondimensional ratings are computed in Table 2.

Now compute the additive weighted scores:

$$S_A = 0.25(0.50) + 0.10(0.50) + 0.15(0) + 0.20(0.80) + 0.30(0.50) = 0.485,$$
$$S_B = 0.25(0) + 0.10(1.00) + 0.15(0.50) + 0.20(1.00) + 0.30(1.00) = 0.675,$$
$$S_C = 0.25(1.00) + 0.10(0) + 0.15(1.00) + 0.20(0) + 0.30(0.50) = 0.550.$$

Therefore, the preferred alternative is Design B.

Table 2: Nondimensional ratings

Attribute	A	B	C
Initial cost	0.50	0.00	1.00
Maintenance	0.50	1.00	0.00
Safety	0.00	0.50	1.00
Reliability	0.80	1.00	0.00
Product quality	0.50	1.00	0.50

6. (15 marks) A U.S. company is considering a high-technology project in a foreign country. The estimated after-tax cash flows for the project, stated in the foreign currency (T-marks), are shown in Table 3.

Table 3: Project cash flows in T-marks (after taxes)

End of Year	Cash Flow (T-marks after taxes)
0	−3,600,000
1	450,000
2	1,500,000
3	1,500,000
4	1,500,000
5	1,500,000
6	1,500,000
7	1,500,000

The analysis period is seven years. The company requires an after-tax rate of return of 18% per year in U.S. dollars on investments made in this country. Assume that the T-mark is expected to devalue relative to the U.S. dollar at an average rate of 12% per year. The current exchange rate is 20 T-marks per U.S. dollar. Answer the following:

(a) (5 marks) Based on a present worth (PW) analysis in U.S. dollars, should the project be approved?

(b) (5 marks) What is the internal rate of return (IRR) of the project in T-marks?

(c) (5 marks) Based on your answer to part (b), what is the IRR of the project in U.S. dollars?

Solution. Part (a). Because the T-mark is expected to devalue by 12%

per year, the equivalent required return in T-marks is

$$(1 + i_T) = (1 + 0.18)(1 + 0.12) = 1.3216,$$

so

$$i_T = 32.16\%.$$

Then

$$PW_{\mathrm{T}} = -3{,}600{,}000 + \frac{450{,}000}{1.3216} + \sum_{t=2}^{7} \frac{1{,}500{,}000}{(1.3216)^t} \approx -392{,}643.31 \text{ T-marks.}$$

Using the current exchange rate of 20 T-marks per U.S. dollar,

$$PW_{\$} = \frac{PW_{\mathrm{T}}}{20} \approx \frac{-392{,}643.31}{20} = -\$19{,}632.17.$$

Since $PW_{\$} < 0$, the project **should not be approved**.

Part (b). The IRR in T-marks, denoted by i^*, satisfies

$$-3{,}600{,}000 + \frac{450{,}000}{(1 + i^*)} + \sum_{t=2}^{7} \frac{1{,}500{,}000}{(1 + i^*)^t} = 0.$$

At $i = 27\%$,

$$PW(27\%) = -3{,}600{,}000 + \frac{450{,}000}{1.27} + \sum_{t=2}^{7} \frac{1{,}500{,}000}{(1.27)^t} \approx +86{,}223.$$

At $i = 28\%$,

$$PW(28\%) = -3{,}600{,}000 + \frac{450{,}000}{1.28} + \sum_{t=2}^{7} \frac{1{,}500{,}000}{(1.28)^t} \approx -14{,}789.$$

Interpolating,

$$IRR_T \approx 27\% + \frac{86{,}223}{86{,}223 + 14{,}789}(1\%) \approx 27.85\%.$$

Therefore, IRR of the project in T-marks is 27.85%.

Part (c). Let $IRR_{\$}$ denote the internal rate of return in U.S. dollars. Since the T-mark is expected to devalue by $f_e = 12\%$ per year,

$$(1 + IRR_T) = (1 + f_e)(1 + IRR_{\$}).$$

Thus,

$$IRR_{\$} = \frac{IRR_T - f_e}{1 + f_e} = \frac{0.1585}{1.12} \approx 0.1415.$$

7. (10 marks) A risk-averse agent has utility function $U(x) = \ln x$ and initial wealth \$60,000. The agent faces a potential loss of \$12,000 with probability 0.1, and no loss with probability 0.9.

   (a) (4 marks) What is the maximum premium the agent would be willing to pay for full insurance against this loss?

   (b) (4 marks) What is the minimum premium that an insurer, with the same utility function $U(x) = \ln x$ and initial wealth \$1,200,000, would be willing to charge to fully insure this loss?

   (c) (2 marks) At what range of premium would the insurer be willing to provide coverage and the agent also be willing to purchase insurance?

Solution. Part (a). Let $p_{\max}$ be the maximum premium the agent is willing to pay.

Without insurance, the agent's expected utility is

$$\mathbb{E}[U(W)] = 0.9 \ln(60{,}000) + 0.1 \ln(48{,}000) \approx 10.9798.$$

With full insurance and premium $p_{\max}$, the agent's wealth is certain and equal to

$$60{,}000 - p_{\max}.$$

At the maximum acceptable premium, the agent is indifferent between buying and not buying insurance, so

$$\ln(60{,}000 - p_{\max}) = 0.9 \ln(60{,}000) + 0.1 \ln(48{,}000),$$

and therefore

$$p_{\max} = 60{,}000 - \exp\Big(0.9 \ln(60{,}000) + 0.1 \ln(48{,}000)\Big) \approx 1{,}324.03.$$

Part (b). Let $p_{\min}$ be the minimum premium the insurer is willing to charge.

If the insurer does not offer coverage, utility is

$$\ln(1{,}200{,}000).$$

If the insurer offers full coverage for premium $p_{\min}$, the insurer's expected utility is

$$0.9(1{,}200{,}000 + p_{\min}) + 0.1(1{,}200{,}000 + p_{\min} - 12{,}000).$$

The insurer is indifferent when

$$\ln(1{,}200{,}000) = 0.9\ln(1{,}200{,}000 + p_{\min}) + 0.1\ln(1{,}200{,}000 + p_{\min} - 12{,}000),$$

and thus

$$p_{\min} \approx 1{,}205.43.$$

Part (c). The range is $[1205.43, 1324.03]$.

8. (8 marks) Consider the cumulative distribution functions of random variables X and Y:

$$F_X(x) = \begin{cases} 0, & x < 2, \\ 0.6, & 2 \le x < 4, \\ 1, & x \ge 4, \end{cases}$$

and

$$F_Y(x) = \begin{cases} 0, & x < 1, \\ 0.5, & 1 \le x < 3, \\ 1, & x \ge 3. \end{cases}$$

Determine:

(a) (4 marks) whether X first-order stochastically dominates Y;

(b) (4 marks) whether X second-order stochastically dominates Y.

Solution. Part (a). X first-order stochastically dominates Y if

$$F_X(x) \le F_Y(x) \qquad \text{for all } x.$$

Note that

$$F_X(x) = 0 < 0.5 = F_Y(x), \qquad 1 \le x < 2,$$

and

$$F_X(x) = 0.6 > 0.5 = F_Y(x), \qquad 2 \le x < 3.$$

The two cumulative distribution functions cross, so X does not first-order stochastically dominate Y.

Part (b). Define

$$D(x) = \int_{-\infty}^{x} \big(F_X(t) - F_Y(t)\big)\, dt.$$

Then

$$D(x) = \begin{cases} 0, & x < 1, \\ -0.5(x-1), & 1 \le x < 2, \\ -0.5 + 0.1(x-2) = 0.1x - 0.7, & 2 \le x < 3, \\ -0.4 - 0.4(x-3) = 0.8 - 0.4x, & 3 \le x < 4, \\ -0.8, & x \ge 4. \end{cases}$$

It follows that

$$D(x) \le 0 \qquad \text{for all } x.$$

Therefore, X second-order stochastically dominates Y.

9. (10 marks) A city is considering purchasing a parcel of land for \$500,000 and constructing an office complex on it. The planning horizon is 20 years. Three mutually exclusive building designs have been proposed by an architectural firm (see Table 4). Using the modified benefit-cost ratio method and a MARR of 10% per year to determine which alternative, if any, should be recommended to the city council.

Table 4: Data for the office-complex alternatives

Item	Design A	Design B	Design C
Initial cost, including land	\$1,300,000	\$1,700,000	\$3,500,000
Resale value of land and building at the end of year 20	\$500,000	\$900,000	\$2,000,000
Annual net rental income (after all operating expenses)	\$120,000	\$300,000	\$450,000

Solution. We first Compute the modified BC ratio for each alternative. For Design A:

$$PW(\text{rent})_A = 120{,}000(P/A, 10\%, 20) = 1{,}021{,}628$$

$$PW(S)_A = 500{,}000(P/F, 10\%, 20) = 74{,}322$$

$$I_A - PW(S)_A = 1{,}300{,}000 - 74{,}322 = 1{,}225{,}678$$

$$(BC)_{m,A} = \frac{1{,}021{,}628}{1{,}225{,}678} = 0.834$$

For Design B:

$$PW(\text{rent})_B = 300{,}000(P/A, 10\%, 20) = 2{,}554{,}069$$

$$PW(S)_B = 900{,}000(P/F, 10\%, 20) = 133{,}779$$

$$I_B - PW(S)_B = 1{,}700{,}000 - 133{,}779 = 1{,}566{,}221$$

$$(BC)_{m,B} = \frac{2{,}554{,}069}{1{,}566{,}221} = 1.631$$

For Design C:

$$PW(\text{rent})_C = 450{,}000(P/A, 10\%, 20) = 3{,}831{,}104$$

$$PW(S)_C = 2{,}000{,}000(P/F, 10\%, 20) = 297{,}287$$

$$I_C - PW(S)_C = 3{,}500{,}000 - 297{,}287 = 3{,}202{,}713$$

$$(BC)_{m,C} = \frac{3{,}831{,}104}{3{,}202{,}713} = 1.196$$

Since the alternatives are mutually exclusive, they must be compared incrementally in order of increasing initial cost:

$$A \;\rightarrow\; B \;\rightarrow\; C.$$

First,

$$(BC)_{m,A} = 0.834 < 1$$

Therefore, Design A is not justified and is rejected, and the do-nothing alternative remains the defender. For Design B,

$$(BC)_{m,B} = 1.631 > 1$$

Therefore, Design B is justified and becomes the current best alternative. Incremental present worth of net rental income:

$$\Delta PW(\text{rent}) = (450{,}000 - 300{,}000)(P/A, 10\%, 20) = 1{,}277{,}035.$$

Incremental net investment:

$$\begin{aligned}\Delta[I - PW(S)] &= (3{,}500{,}000 - 1{,}700{,}000) - (2{,}000{,}000 - 900{,}000)(P/F, 10\%, 20) \\ &= 1{,}636{,}492\end{aligned}$$

Thus, the incremental modified BC ratio is

$$(\Delta BC)_m = \frac{1{,}277{,}035}{1{,}636{,}492} = 0.780 < 1.$$

The additional investment required for Design C is not justified. Therefore, the preferred alternative is Design B.